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OUR OBJECTIVE

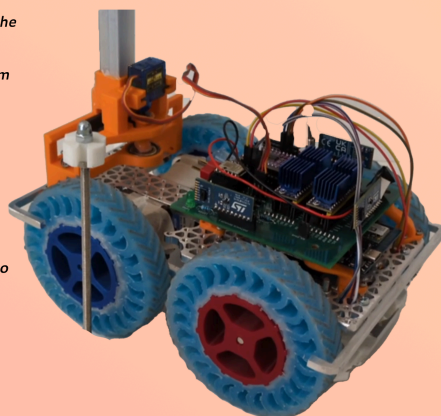
We are Tachyons, a team founded in November 2025 in Giarre, Sicily.

The primary objective of our team is to deploy a highly reliable and innovative Rescue Maze robot. The architecture is engineered to autonomously explore the maze, detect walls, classify floor tiles, dynamically update an internal map, identify victim targets, and execute precise rescue-kit deployment. From a technical perspective, our goals focus on three pillars:

1) **Sensor Matrix Optimization:** Maximizing the efficiency of our 8 Time-of-Flight (ToF) distance matrix sensors for precise wall detection and spatial mapping.

2) **Mechatronic Reliability:** Transitioning from rapid-prototyping materials (PLA and perboards) to industrial-grade hardware to ensure structural integrity and eliminate electrical or thermal points of failure.

3) **Algorithmic Adaptability:** Implementing robust maze-solving, calibration routines, and visual recognition systems that adapt seamlessly to varying international competition arena environments.



QB8

TECHNICAL DESCRIPTION

The robot utilizes an Arduino Uno Q to achieve precise control of the stepper motors via a PID loop. It features custom, airless silicone wheels manufactured through injection molding inside 3D-printed ABS molds. On the mechanical side, it incorporates several in-house developed solutions: custom wheels, a suspension system, a compliant rescue kit launcher, a microswitch-equipped bumper, and a CNC-machined aluminum isogrid chassis designed for modularity and flexibility.

4x TMC2209 Stepper Drivers: Used to control our stepper motors.

They are efficient and enable smooth motor rotation with minimal vibration while also having bi-directional UART communication which allows us to change settings during the run and read motor loads. They can handle up to 256 microsteps

1x PCA9548A I2C Multiplexer: Used to connect the distance and color sensors to a single I2C port on the Arduino Uno Q.

3x VL53L7CX Time-of-Flight (ToF) Ranging Sensor:

A state-of-the-art FlightSense multi-zone distance sensor capable of delivering an 8*8 spatial grid array. Utilizing an invisible 940nm VCSEL laser, it measures distances up to 350cm across 64 distinct zones simultaneously.

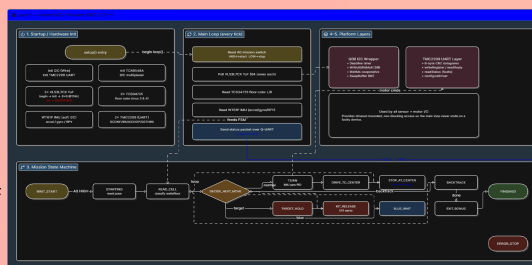
1x WT61P 6-Axis IMU: Integrates a 3-axis gyroscope, accelerometer, and an onboard Kalman filter to deliver high-precision angle data. It allows exact 90 degrees turns and reliable odometry calibration.

2x TCS3472 Color Sensor: RGBC photodiode array with IR-blocking filter for accurate tile color detection.



SOFTWARE

Q-Botto software implements strict functional isolation between autonomous operation and operator support functions. Mission-critical functionality is handled by the Arduino UNO Q MCU that manages low-level sensor input, real-time motor control, safety modes, and Dijkstra-based frontier exploration. It provides autonomous run robustness even in case of failure of secondary components. Operator support tasks are delegated to an MPU-side Python application, communicated with through Q-UART protocol. In order to achieve maximum FPS, MPU runs a light-weight vision pipeline: two USB cameras provide video data that are processed using OpenCV to get ROIs. In lieu of full frame processing, victim detection uses cropped frames that are analyzed through an ONNX model trained for Greek letters identification whereas cognition analysis uses native OpenCV ring analysis.



MEET THE SQUAD



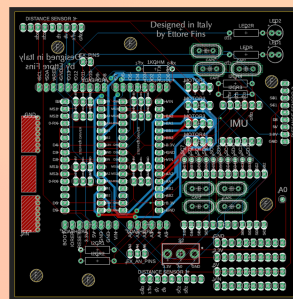
ETTORE FINIS: Team lead, Hardware and mechanics engineer and developer, Electronics and PCB designer and engineer, Software engineer, technical documentation author, test engineer. In 2025, he served as a key member of the MezaMaze rescue maze team, where he reached first place in the regionals and third in the national and European competitions. This year, he decided to form his own team to focus on the development of even more advanced hardware. He has designed all the electronics of the robot. He is a main part of the software development, and he designed the CAD of the current version of the robot and manufactured all the custom parts.

FRANCESCO SCIUTO: Software developer, mapping algorithm author. In 2025, he served as the lead software developer and graphic designer for an On Stage Advanced team (Madagabot), which was awarded Best Technical Poster at the National Qualifiers and reached the European RoboCup Junior Championship. Building on this success, for the 2026 season, he transitioned his expertise into the Rescue Maze category, focusing on the MCU-side mission layer, including autonomous navigation, maze-mapping logic, and low-level sensor integration.

GABRIEL GRASSO: Test and hardware engineer, woodworker, and labyrinth maker. This is the first time that he participates in Robocup Jr. He worked on building a wooden maze for testing purposes, and he designed some previous versions of the robot in CAD

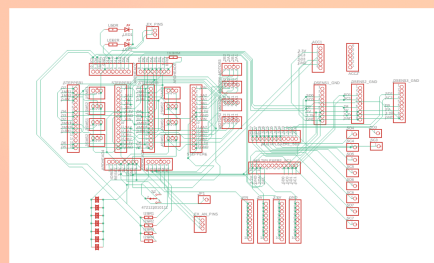
ANGELO CARUSO: Graphic designer, technical documentation author. In 2025, he was part of an On Stage entry team (FermiTuttiInT), where he was the main software developer and a jack of all trades for the construction of his main robot. For the 2026 season, he focused on the MPU-side software infrastructure, taking charge of the high-level Python dashboard, vision diagnostics, and the implementation of the embedded machine learning models for target recognition.

CUSTOM MADE PCB

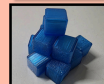


The PCB has four layers. The outer layers are used for signal traces and the inner layers are for large polygons, one connected to GND, and one connected to VIN in order to minimize interference and inductance. On the left is an image of the traces of layers 1 and 4, and below is an image of the electrical diagram. The PCB has all the electronic components of the robot, except for the color sensors, soldered, or, for the less reliable components, held by pinheaders for quick and easy replacement.

ELECTRICAL DIAGRAM



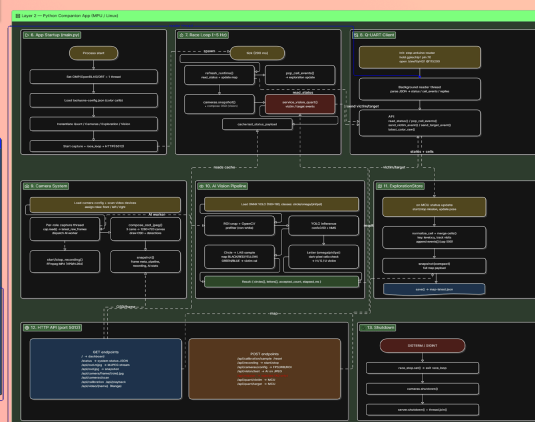
INNOVATIVE HARDWARE SOLUTIONS



To address traction issues, we developed custom silicone wheels manufactured via injection molding inside an inert, 3D-printed ABS mold, which reduces adhesion and allows for complex tread geometries that clear obstacles and minimize dirt buildup. To reduce weight by 60% while increasing softness and shock absorption, the wheels incorporate internal TPU inserts that create a cavity within the silicone.

Because custom wheels alone were insufficient, we also developed a rear suspension system based on the historic Christie tank design. The stepper motor for each front wheel is mounted on a bearing-connected lever, allowing it to pivot vertically with the assistance of a return spring, which enables the robot to clear 3 cm obstacles easily.

Drawing from previous Rescue Maze experience, we engineered custom rescue cubes using a low-coefficient-of-restitution polymer combined with an internal energy dissipation mechanism to minimize bouncing. This stability allowed the creation of an advanced deployment system that "shoots" the kits so they land directly in front of the victims. Finally, for navigation and collision handling, the robot utilizes two microswitches mounted inside a compliant bumper to detect and distinguish whether impacts occur on the right, left, or center.



SiciliaFiera, Catania 18/04/2026
Nationals



4th place
Regionals

2nd place
National